Xinyi (Evelyn) Gao

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Education

University of Chicago — **Expected June 2028**
Bachelor of Science in Computer Science and Bachelor of Arts in Statistics (GPA: 3.76/4.0) — *Chicago, IL*

Technical Skills

Languages: Python, SQL, R, Java, C, TypeScript, JavaScript, HTML/CSS
Machine Learning & Data: scikit-learn, Pandas, NumPy, SciPy, GeoPandas, Matplotlib, feature engineering, PCA, UMAP, clustering, time-series forecasting, semantic search & reranking
AI Engineering: Model Context Protocol (MCP), FastMCP, LangChain, OpenAI Embeddings, agentic orchestration, LLM evaluation harnesses, prompt-driven tool design
Tools & Frameworks: Git, GitHub Actions (CI/CD), pytest, Docker, Pydantic, Supabase, React, Next.js

Experience

Salk Institute, Harnessing Plants Initiative — **Jun. 2026 – Aug. 2026**
Research Software Engineering Intern, PIs: Talmo Pereira & Wolfgang Busch — *La Jolla, CA*

- Shipped bloom-mcp, a production agentic orchestration platform exposing 22 validated tools via FastMCP and Pydantic, letting Claude and other LLMs plan and execute self-validating root-phenotyping workflows – QC, outlier detection with automatic fallback, scikit-learn PCA/clustering, UMAP, and cross-experiment statistics – from plain-English requests.
- Architected a reproducibility and observability contract layer for non-deterministic agent runs – validated every tool's inputs and outputs, recording random seeds, library versions, and SHA-256 output hashes for deterministic execution.
- Engineered pluggable persistence across Supabase, S3, and local private-data deployments, balancing latency, cost, and reliability; replaced 649–880 sequential database round trips with a single bulk-read RPC.
- Built the platform's evaluation harness: reproduced published wheat analyses as golden-value regression tests (85 test files, 1,211 tests, 22 golden fixtures); delivered all 22 tools test-first behind CI quality gates and peer-reviewed PRs.

Sony — **May 2025 – Sep. 2025**
Computer Science Operations & Maintenance Intern — *Beijing, China*

- Built production Python delivery pipelines handling 100+ weekly releases and assets – retrieval, metadata cleaning, batch processing, and dispatch – cutting release preparation by ~60% and eliminating manual-detail errors.
- Designed a config-driven keyword- and regex-based classifier routing 300+ user feedback comments weekly in seconds; YAML-only updates enabled reliable pattern changes without code deployments.
- Established safe change management for a 100,000+ row production database: validated SQL updates on non-production replicas and applied batched changes with per-batch row-count and sample verification.

Tsinghua University, Vehicle Emission Research Group — **Jul. 2024 – Jan. 2025**
Research Assistant, Spatiotemporal Modeling and Forecasting — *Beijing, China*

- Built a leakage-resistant pipeline forecasting vehicle emissions 24 hours ahead from 30 days of time-series data, using lagged and seasonal features with time-based splits; rigorously evaluated Random Forest against an ARIMA baseline using MAE.
- Detected statistically significant emission hotspots with Moran's I and Getis-Ord Gi* in GeoPandas, prioritizing regions for potential regulatory investigation.

Projects

Proxima | *OpenAI Embeddings, Vector Similarity, Python, Next.js, TypeScript* — **Jan. 2026 – Jun. 2026**

- Built Proxima, a production semantic-search product matching students with research professors: a two-stage orchestration pipeline recalling the top 10 of 1,500+ OpenAI profile embeddings by cosine similarity, then reranking to a final top 3 with GPT-4o alongside personalized match explanations.
- Kept ranking reliable during external API outages via prompt engineering and a keyword fallback; evaluated non-deterministic LLM ranking behavior against hand-labeled matches – 87% top-three relevance, 97% coverage, and 89% best-choice accuracy.
- Surfaced per-match confidence scores and React relevance visualizations, improving observability and user trust.

AlmaBot | *Python, LangChain, Gradio* — **May 2025 – Present**

- Built a graph-based course-planning engine using prerequisite dependency graphs, Kahn's algorithm, topological sorting, and cycle detection to sequence degree requirements across terms and surface unsatisfiable prerequisite chains.
- Built a stateless, client-only Gradio tool indexing 4,502 course records across 187 subjects/majors; used LangChain to translate preferences into prerequisite, workload, Gen Ed, and timed-section constraints.